

Decision and Risk Analysis

Lecture 1: **Bayesian Thinking for Risk and Decision Analysis**

DAEN 427/ISEN 427/ISEN 627

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A decision problem begins with a **choice to be made** under certain **objectives** and **constraints**.

Risk and Uncertainty

- **Choice under risk:** Outcome probabilities are known (or can be estimated reliably).
- **Choice under uncertainty:** Outcome probabilities are unknown (or difficult to estimate).

Decision and Risk Analysis

1. **Modeling Decisions:** Define the available choices or actions.
2. **Modeling Risk:** Analyze situations where outcome probabilities are known.
3. **Modeling Uncertainty:** Analyze situations where outcome probabilities are unknown (or difficult to estimate).
4. **Modeling Preferences:** Represent how decision makers value outcomes and trade-offs.

Decision and Risk Analysis, Examples

1. **Modeling Decisions**

A student chooses between studying for an exam tonight or attending a social event.

2. **Modeling Risk**

A company launches a new product knowing there is a 70% chance of high demand and a 30% chance of low demand.

3. **Modeling Uncertainty**

A farmer decides what crop to plant without knowing how much rain will fall this season.

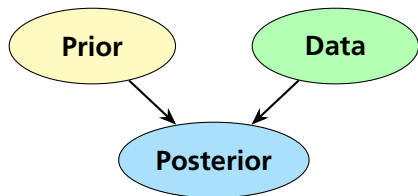
4. **Modeling Preferences**

A traveler chooses between a cheaper flight with a long layover and a more expensive direct flight.

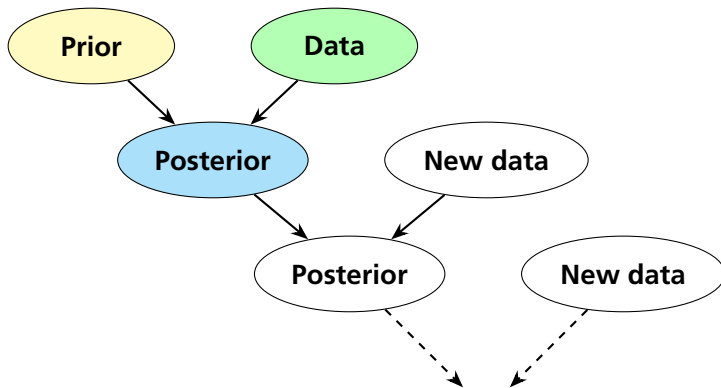
Bayesian Modeling for Decision Analysis

- Bayesian modeling provides a way to update beliefs as new information becomes available.
- It combines:
 1. **Prior beliefs**: what we know / assume before seeing new data.
 2. **Evidence/data**: new information relevant to the decision.
 3. **Posterior beliefs**: updated probabilities after incorporating the evidence.
- In decision analysis, Bayesian modeling helps decision makers revise risk estimates and make better choices under uncertainty.
- In short, it supports learning from evidence and improving decisions over time.

A Bayesian Knowledge-Building Diagram



A Bayesian Knowledge-Building Diagram



Quiz 1

When flipping a fair coin, we say that “the probability of flipping Heads is 0.5.” How do you interpret this probability?

- a. If I flip this coin over and over, roughly 50% will be Heads.
- b. Heads and Tails are equally plausible.
- c. Both a and b make sense.

Quiz 2

An election is coming up and a pollster claims that candidate A has a 0.9 probability of winning. How do you interpret this probability?

- a. If we observe the election over and over, candidate A will win roughly 90% of the time.
- b. Candidate A is much more likely to win than to lose.
- c. The pollster's calculation is wrong. Candidate A will either win or lose, thus their probability of winning can only be 0 or 1.

Quiz 3

Consider two claims:

- 1 Zuofu claims that he can predict the outcome of a coin flip. To test his claim, you flip a fair coin 10 times and he correctly predicts all 10.
- 2 Kavya claims that she can distinguish natural and artificial sweeteners. To test her claim, you give her 10 sweetener samples and she correctly identifies each.

In light of these experiments, what do you conclude?

- a. You're more confident in Kavya's claim than Zuofu's claim.
- b. The evidence supporting Zuofu's claim is just as strong as the evidence supporting Kavya's claim.

Quiz 4

Suppose that during a recent doctor's visit, you tested positive for a very rare disease. If you only get to ask the doctor one question, which would it be?

- a. What's the chance that I actually have the disease?
- b. If in fact I don't have the disease, what's the chance that I would've gotten this positive test result?

Interpreting Probability

- The frequentist philosophy is so named for its interpretation of probability as the long-run relative frequency of a repeatable event.
- In the Bayesian philosophy, a probability measures the relative plausibility of an event.

Is this coin fair?

- **Frequentist:** "If the coin were fair, how surprising would these flips be?"
- **Bayesian:** "Given these flips, how much should I believe the coin is fair?"

One-Time Events

- Many important events are unrepeatable:
Big Bang, elections, weather and climate change, ...
- **Frequentist** probability can be awkward for one-time events because there is no long-run repetition.
- **Bayesian** probability is more flexible:
 - It expresses degree of belief.
 - It incorporates model, evidence, and uncertainty.
- A 0.9 probability of winning means the candidate is considered 9 times more likely to win than lose.

The Bayesian Balancing Act

- Bayesian and frequentist approaches interpret the same evidence differently:
 - **Frequentist:** Treats "10 out of 10" identically in all cases.
 - **Bayesian:** Plausibility on observed evidence given prior beliefs.

Both achieve 10 out of 10 correct in testing:

- Zuofu claims he can predict coin flips.
... is doubtful, predicting coin flips is highly implausible.
- Kavya claims she can distinguish natural / artificial sweeteners.
... is more believable.

Ask Questions!

- A frequentist analysis assesses the uncertainty of the observed data in light of an assumed hypothesis.
- A Bayesian analysis assesses the uncertainty of the hypothesis in light of the observed data.

Scenario:

- Population: 100 people
- 4 have the disease, 96 do not have the disease

	Test +	Test –	Total
Disease	3	1	4
No disease	9	87	96
Total	12	88	100

Frequentist question: $P(\text{Positive Test} \mid \text{No Disease})$

- Among the 96 healthy people, 9 tested positive
- False positive rate: $\frac{9}{96} \approx 0.10$
- If a person does *not* have the disease, there is about a 10% chance of testing positive.
 - Frequentist analysis focuses on: $P(\text{Data} \mid \text{Null Hypothesis})$
 - A p-value is *not* the probability that the hypothesis is true.
(common misunderstanding)

Bayesian question: $P(\text{Disease} \mid \text{Positive Test})$

- Only 3 of the 12 positive tests are true positives.
- Probability of actually having the disease: $\frac{3}{12} = 0.25$
- Even with a positive test, there is only a 25% chance of having the disease.
 - Sensitivity : $P(+ \mid \text{Disease}) = \frac{3}{4} = 75\%$
 - Specificity : $P(- \mid \text{No Disease}) = \frac{87}{96} \approx 90.6\%$

The Meaning of Probability

Probability describes uncertainty.

Frequentist:

- Probability is a **long-run frequency**.
- Describes what happens over many repeated trials.

Bayesian:

- Probability is a **degree of belief**.
- Measures how plausible something is given current information.

Bayesian thinking updates beliefs using evidence:

$$\text{Posterior} \propto \text{Prior} \times \text{Likelihood}$$

Probability is not only about randomness;
it is a mathematical language for learning under uncertainty.

Homework

Question 1

(When was the last time you changed your mind?)

Think of a recent situation in which you changed your mind. As with the Italian restaurant example (Figure "Bayesian Knowledge-Building Diagram"), make a diagram that includes your prior information, your new data that helped you change your mind, and your posterior conclusion.

Homework

Question 2 (Applying for an internship)

There are several data scientist openings at a much-ballyhooed company. Having read the job description, you know for a fact that you are qualified for the position: this is your data. Your goal is to ascertain whether you will actually be offered a position: this is your hypothesis.

- a. From the perspective of someone using frequentist thinking, what question is answered in testing the hypothesis that you'll be offered the position?
- b. Repeat part a from the perspective of someone using Bayesian thinking.
- c. Which question would you rather have the answer to: the frequentist or the Bayesian? Explain your reasoning.

Homework

Question 3 (You already know stuff)

There are several data scientist openings at a much-ballyhooed company. Having read the job description, you know for a fact that you are qualified for the position: this is your data. Your goal is to ascertain whether you will actually be offered a position: this is your hypothesis.

- a. Identify a topic that you know about (e.g., a sport, a school subject, music).
- b. Identify a hypothesis about this subject.
- c. How would your current expertise inform your conclusion about this hypothesis?
- d. Which framework are you employing here, Bayesian or frequentist?

Homework

Question 4 (Benefits of Bayesian Statistics)

Your friend just became interested in Bayesian statistics. Explain the following to them:

- a. Why is Bayesian statistics useful?
- b. What are the similarities in Bayesian and frequentist statistics?

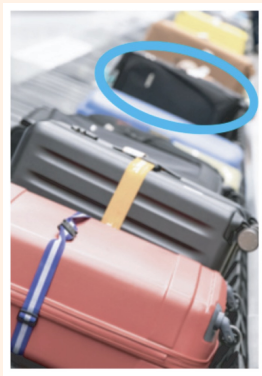
Question 5

Let's suppose that you are an air traveler along with 99 other passengers from your flight, each of whom, like you, checked one item of luggage.

A recording piped through the speakers reminds you that: *"Many bags look alike. Please check your bag carefully before exiting the terminal."*

Indeed, your bag is one of the most popular models on the market, a medium-sized black rectangular case used by 5 percent of all travelers.

Given your distance from the luggage chute, you can discern only the shape, size, and color of the bags that emerge, not any detailed markings on the bags.



[5a.] Let's suppose that the first bag from your flight to enter the luggage carousel indeed has the same shape, size, and color as yours. Is it yours? Explain (i.e., calculate the posterior probability that the 1st bag is yours).

[5b.] Now suppose that we continue to wait for our bag to appear, failing to see it among the first 98 bags to enter the carousel (i.e., only 2 bags are left to come off the airplane). Calculate the posterior probability that the 99th bag, which also matches ours in shape, size, and color, is our own.

[a.] Let's suppose that the first bag from your flight to enter the luggage carousel indeed has the same shape, size, and color as yours. Is it yours? Explain (i.e., calculate the posterior probability that the 1st bag is yours).

Answer: Let Y = first bag is yours, and M = first bag matches.

$$P(Y) = \frac{1}{100}$$

$$\mathcal{L}(Y^c | M) = P(M | Y^c) = 0.05$$

$$\mathcal{L}(Y | M) = P(M | Y) = 1$$

$$\begin{aligned} P(Y | M) &= \frac{P(M | Y)P(Y)}{P(M)} = \frac{P(M | Y)P(Y)}{P(M | Y)P(Y) + P(M | Y^c)P(Y^c)} = \\ &= \frac{1 \cdot 0.01}{1 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.01}{0.0595} \approx 0.168 \end{aligned}$$

So even though the first bag matches, it is probably not yours.

[b.] Now suppose that we continue to wait for our bag to appear, failing to see it among the first 98 bags to enter the carousel (i.e., only 2 bags are left to come off the airplane). Calculate the posterior probability that the 99th bag, which also matches ours in shape, size, and color, is our own.

Answer:

$$\begin{aligned} P(Y \mid M) &= \frac{1 \cdot \frac{1}{\text{total of bags remaining}}}{1 \cdot \frac{1}{\text{total of bags remaining}} + 0.05 \cdot \frac{\text{total of bags remaining} - 1}{\text{total of bags remaining}}} \\ &\quad \text{(if 2 bags remain)} \\ &= \frac{1 \cdot \frac{1}{2}}{1 \cdot \frac{1}{2} + 0.05 \cdot \frac{1}{2}} = \frac{1}{1 + 0.05} \approx 0.9524 \end{aligned}$$

There is a 95.24% chance that the 99th bag is mine given that it looks like mine.